

# Does an LLM design better quantum circuits, or just bigger ones?

A five-gate, fixed-length comparison of LLM-guided and classical circuit search

**Headline:** last week's LLM advantage was mostly a circuit-size decision. Remove that one degree of freedom and it disappears on three of four tasks.

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## WHAT CHANGED

# Last week's search let the LLM choose the circuit size

### Last week (2026-07-24 pilot)

- Circuit body: 1 to 5 gates - length was itself a choice
- Budget B = 4 candidates, 2 seeds
- Reported: LLM open-loop 0.155 vs Random 0.189
- Its own config file said: "integration/smoke demonstration only; not a statistically powered result"

### This week

- Circuit body: EXACTLY 5 gates - length is fixed for every method
- Budget B = 8 candidates, 10 seeds
- Five arms instead of three
- Three conditions run so the difference can be attributed, not guessed

**Why this matters:** if circuit length is a free variable and longer circuits are usually better, then any proposer that simply always asks for the maximum length wins - without designing anything.

# Every rule, stated so you can check any circuit by eye

| Constraint           | Value                                                                                |
|----------------------|--------------------------------------------------------------------------------------|
| Qubits               | 3 for T1 and T2 ( $2^3 = 8$ amplitudes); 5 for T3 and T4 ( $2^5 = 32$ )              |
| Gate count           | EXACTLY 5 - not a maximum, not a range. Identical for every method.                  |
| Gate set (7 types)   | one wire: H, RX, RY, RZ    two wires: CRX, CRY, CRZ                                  |
| Angles               | one continuous theta per gate in $[-\pi, \pi]$ ; H carries none                      |
| Training             | NONE. The proposed angles are evaluated verbatim - no optimizer exists on this path. |
| Encoding / readout   | amplitude encoding; Pauli-Z on qubit 0; prediction = $(1 - \langle Z_0 \rangle)/2$   |
| Classical parameters | zero - no dense layer can rescue a weak circuit                                      |

**One operation = one physical gate here.** There are no layer macros: a 5-gate circuit compiles to 5 gates, so "5" means the same thing to every arm and to the reader.

# One number, defined once: RMSE

- Validation RMSE drives the search. Each arm evaluates its candidates and keeps the best one by validation RMSE.
- Test RMSE is reported. The held-out split is touched once per cell, after the choice is frozen, so the reported number was never used to make that choice.
- T1-T3 regress a continuous quantity, so RMSE is the natural metric.
- T4 is binary classification. RMSE against a 0/1 label equals the square root of the Brier score - a proper scoring rule - so all four panels share one axis. AUROC is reported alongside it, and slide 7 shows why that matters.

**RMSE is comparable between arms within one task, never between tasks.** T1, T2, T3 and T4 predict different quantities on different scales, so the panels each carry their own y-axis and a T1 value must not be read against a T3 value.

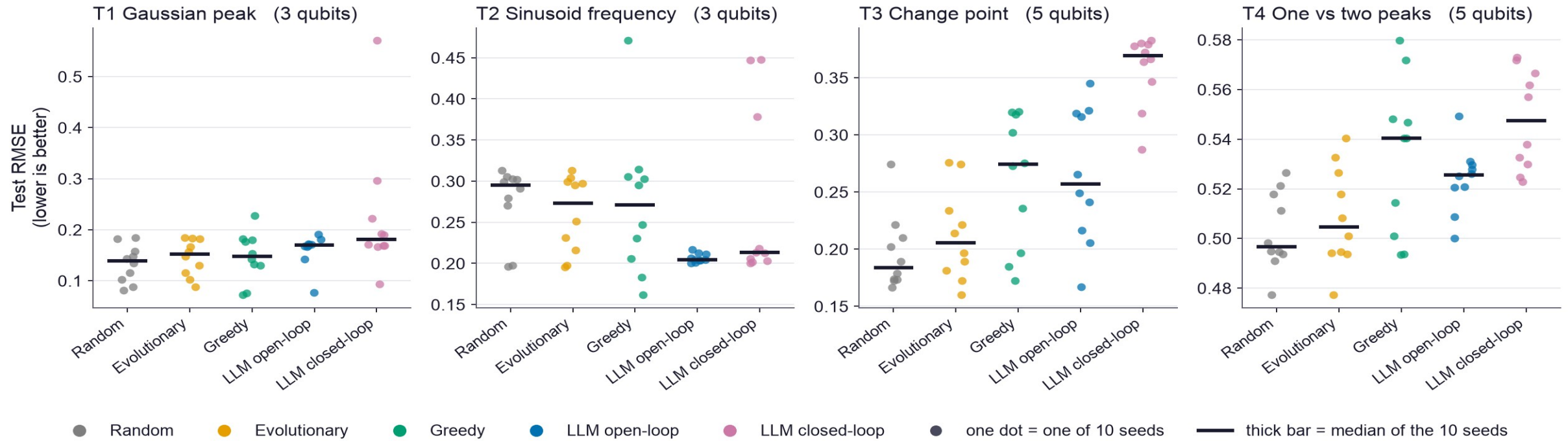
# Five arms, one shared budget of unique candidates

| Arm             | How it proposes                                                           | API calls       |
|-----------------|---------------------------------------------------------------------------|-----------------|
| Random          | draws 5 gates, wires and angles uniformly                                 | none            |
| Evolutionary    | ( $\mu + \lambda$ ), $\mu = 4$ : seed 4 at random, then mutate a survivor | none            |
| Greedy          | hill climbing: mutate the incumbent, keep it only if validation improves  | none            |
| LLM open-loop   | one request per candidate; sees only the model contract and n             | 1 per candidate |
| LLM closed-loop | same, plus validation-only results for what it already tried              | 1 per candidate |

**B = 8 unique evaluated candidates per arm, 10 seeds.** A duplicate or an ungrammatical proposal is recorded but consumes no budget, so no method can buy extra evaluations by proposing badly - and no method gets fewer because it proposes well. One candidate per request, so the closed loop is genuinely sequential: propose, see the result, propose again, eight times.

## RESULT

# Random search wins on three of four tasks



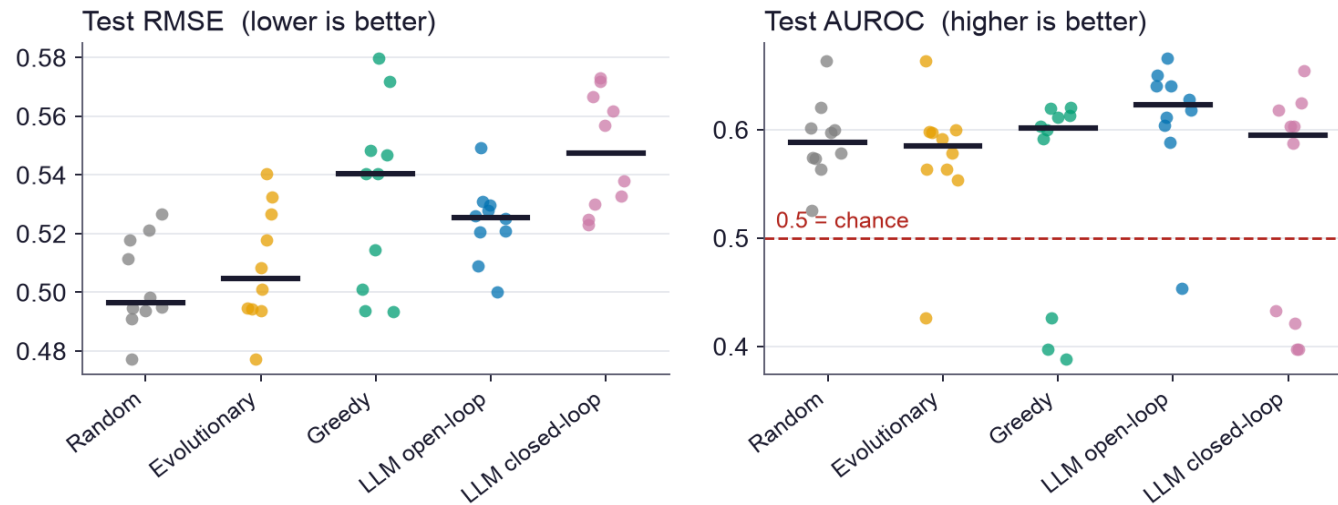
Each panel has its own y-axis: RMSE is comparable between arms within a panel, never between panels, because the tasks regress different quantities.

**The LLM wins on T2 alone, and there it wins clearly (0.204 vs 0.295). Closed-loop never beats open-loop on any task: eight rounds of feedback did not help this model at this budget.**

## METRIC CHECK

# On T4 the metric, not the method, picks the winner

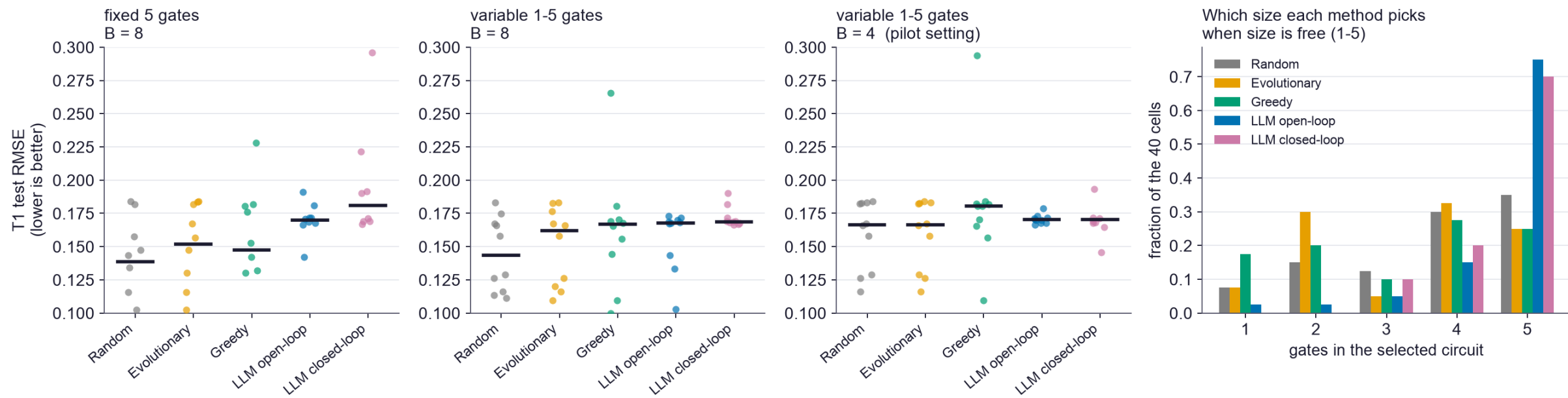
T4 is binary classification: RMSE against a 0/1 label is  $\sqrt{\text{Brier}}$ , a proper score, but it mixes calibration with discrimination.



- By RMSE, Random leads. By AUROC, LLM open-loop leads. Same predictions, opposite ranking.
- RMSE punishes a circuit that ranks the two classes correctly but outputs 0.45 and 0.55 instead of 0 and 1. AUROC does not.
- With five gates and no optimizer the outputs sit near 0.5, so this failure mode is live here, not hypothetical.
- Both numbers are shown because either one alone asserts a different answer.

**Neither reading is strong.** AUROC spans roughly 0.59 to 0.62 against 0.5 for chance. The honest summary of T4 is that five gates without training barely solve it at all, and that no arm separates convincingly.

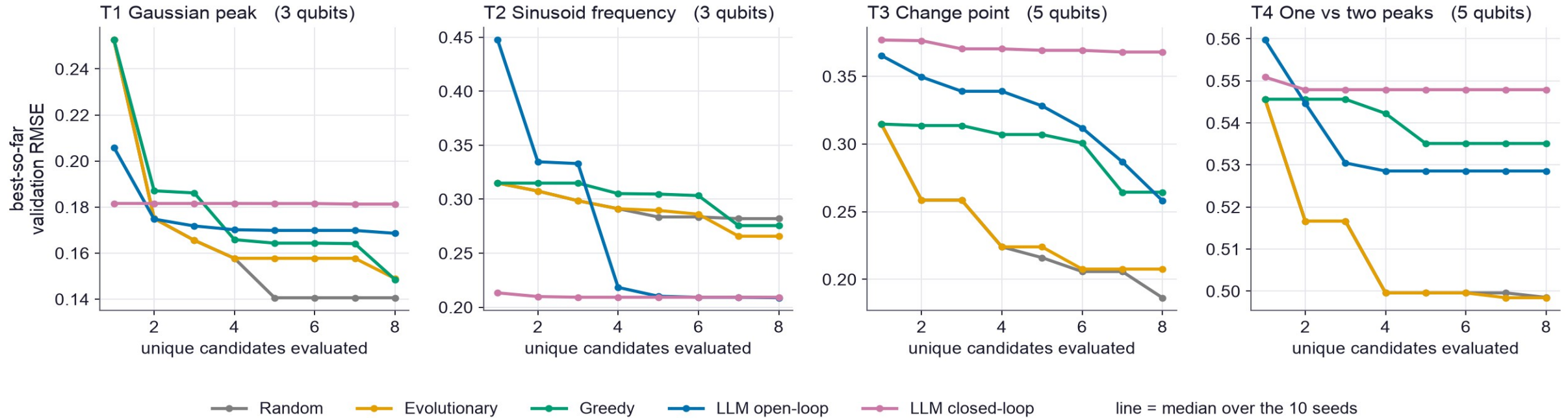
## Last week's advantage was a size decision, not a design one



Left three panels: one dot = one of 10 seeds, thick bar = median. Only Random degrades as circuit size becomes a free choice, because the LLM almost always picks the maximum size while Random draws it uniformly. The pilot rewarded that single decision.

**Right panel:** when length is free the LLM picks the maximum in about 7 of 10 cells while the classical arms spread over 1 to 5. **Left three:** only Random degrades as length is freed and the budget shrinks (T1 median 0.139 → 0.167), while the LLM barely moves (0.170 → 0.170). At the pilot's setting they meet.

## More feedback did not translate into better circuits



- Closed-loop received eight rounds of validation-only feedback on its own candidates and still did not beat open-loop on any task.
- A first attempt measured a 74% duplicate rate for closed-loop, with 11 of 40 cells starved of budget. The prompt now lists every circuit already tried; duplicates fell to 16% and every cell reached its full budget. The numbers here are from the corrected run.

# What I can and cannot claim

## Supported by this run

- With circuit length fixed, gpt-5-nano does not beat random search on 3 of 4 tasks
- It does beat every classical arm on T2, clearly and across all 10 seeds
- When length is free it picks the maximum ~70% of the time; uniform sampling does not
- That single decision explains most of last week's reported advantage

## NOT supported

- "LLMs cannot design quantum circuits" - one small model, one budget, four tiny tasks
- "Closed-loop feedback does not work" - it failed here at 8 rounds on 5-gate circuits
- Any claim about hardware: this is noiseless statevector simulation throughout
- Any cross-task comparison of RMSE values

**Next:** give the LLM a decision that is actually hard. Fixed length removed the easy win; the open question is whether a larger budget, a stronger model, or a task where structure matters more can produce an advantage that survives this kind of attribution check.